Answers

Exercise 1.1

(1) (i) $\{2, 3, 5, 7\}$ (ii) $\{1\}$ (iii) $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ (iv) $\{-5\}$

(2) $\{x \in \mathbb{R} : x^2 = 1\}$

(3) (i) finite (ii) infinite (iii) infinite (iv) infinite (v) infinite

(5) not true (6) 0 (7) 128 (8) $\{0, 1, 2, 3\}$

(9) $A = \{x, y, z\}$ and $B = \{1, 2\}$ (10) $\{(-1, 0), (-1, 1), (0, 2), (1, 2)\}$

Exercise 1.2

(1) (i) reflexive; not symmetric; transitive (ii) not reflexive; symmetric; not transitive

(iii) reflexive; not symmetric ;not transitive (iv) reflexive; symmetric

(v) R is an empty set; not reflexive; symmetric; transitive

(2) (i) (c, c) and (d, d) (ii) (c, a)

(iii) nothing to include (iv) $(c, c), (d, d), (c, a)$ to be included

(3) (i) (c, c) (ii) (c, a) (iii) nothing (iv) (c, c) and (c, a)

(5) $\{(3, 8), (6, 6), (9, 4), (12, 2)\}$; not reflexive; not symmetric; transitive; not an equivalence relation

(7) $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (4, 1), (4, 2), (5, 1)\}$; not reflexive; symmetric; not transitive; not equivalence

(8) smallest set is $\{(a, a), (b, b), (c, c)\}$; largest set is $A \times A$

Exercise 1.3

(1) Yes; inverse is not a function

(2) $f(-4) = 8,\ f(1) = 0,\ f(-2) = 6,\ f(7) = 0,\ f(0) = 0$

(3) $f(-3) = 1,\ f(5) = 38,\ f(2) = 1,\ f(-1) = -5,\ f(0) = -3$

(4) (i) a function; not one–to–one and not onto (ii) not a function

(5) (i) $\{(1, a), (2, a), (3, a), (4, a)\}$ (ii) not possible (iii) not possible

(iv) $\{(1, a), (2, b), (3, c), (4, d)\}$

(6) $\mathbb{R} - \{n\pi + (-1)^n \frac{\pi}{6}\}, n \in \mathbb{Z}$ (7) $\emptyset$

(8) $(-\infty, -\frac{1}{3}] \cup [1, \infty)$ (9) $\mathbb{R} - \{0\}, \mathbb{R} - \{0\}$

(10) for all x, $(g \circ f) = 0$; for $x < 0$, $f \circ g = -4x$; for $x \geq 0$, $f \circ g = 0$.

(12) $f^{-1}(x) = \frac{x + 5}{3}$ (13) $x > 0$

(15) total cost $= 0.43m + 50$, airfare = ₹738

(16) $(A + S)(x) = 55,000 + 0.09x$, total family income = ₹14, 05, 000